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Chandkheda, Ahmedabad
Affiliated

VISHWAKARMA GOVERNMENT
ENGINEERING COLLEGE
CHANDKHEDA, AHMEDABAD.

A
Project Report
On:

Gesture Recognition For Control Systems

Under subject of
DESIGN ENGINEERING – 1A
B.E. III, SEMESTER III
(CSE-Data Science Branch)

SUBMITTED BY:
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Prof. KARAN.P.BHATT
(Faculty Guide)
Academic Year
(2024-2025)

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1) INTRODUCTION

1.1 Introduction about Team Members

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1.2 Introduction to faculty guide

Prof. KARAN.P.BHATT
Assistant professor in Vishwkarma Government Engineerig College.

1.3 Understanding about Design Engineering

Design Engineering is based on globally accepted Design Thinking methodology.

Design is a plan of a system, its implementation and utilization for attaining for a goal. It is to change undesired situation into desired situation means to find solution undesired/uncomfortable situation

1.3 Identification Of Component For Design Thinking

Design thinking is dividing into 6 factors. The 6 six factors are These factors are very useful in design thinking and for study design engineering.

- Observation
- Empathy
- Ideation
- Product Development
- Prototype
- Test

Team building is important in design engineering so that you share your thought to your team member and also share their thoughts and with this thinking you make something unique which is useful in real life. In design thinking, we also take care about social economy so local area's people also use it. Nowadays there is huge project scope of design thinking.

Our journey through this project is useful for us and people who will use this project. Design thinking is that you want to make something that you think and apply it in life so by it your life in that way is much easier than it was.

2) INTRODUCTION TO GESTURE RECOGNITION FOR CONTROL SYSTEM

Gesture recognition is an innovative technology that interprets human gestures as inputs for controlling systems, enabling a more intuitive and natural form of human-computer interaction. In the context of a **Design Engineering project**, gesture recognition can be applied to various control systems, such as home automation, robotic systems, gaming interfaces, or industrial controls, by allowing users to interact with devices using simple hand or body movements instead of traditional input methods like keyboards or touchscreens.

What is Gesture Recognition?

Gesture recognition involves capturing human gestures using sensors, cameras, or specialized hardware (such as motion detectors or accelerometers) and translating those gestures into commands that a system can understand. These gestures could include movements of the hands, arms, or even the entire body, depending on the system's requirements. Modern systems use algorithms based on computer vision and machine learning to accurately identify and interpret gestures in real-time.

Applications in Control Systems

In a control system, gesture recognition can enhance user experience by allowing for hands-free operation and improving accessibility, especially in environments where traditional input devices might not be feasible. For example, gestures could be used to:

- **Control robotic arms** in manufacturing without physical contact.
- **Navigate through presentations** during meetings or conferences.
- **Control smart home devices** such as lighting, temperature, and security systems.
- **Interact with medical equipment** in sterile environments like operating rooms, where touch-based controls may introduce risks.

3) EMPATHY CANVAS

3.1 Selected Users and its Stakeholders

Users

- Hazardous environment
- Factory worker
- Research developer
- Gamers audience

Stakeholder

- System investors
- Services provider
- Medical manufactures
- End users

3.2 Activities of Users

- Model training
- Environment Testing
- Design improvement
- User feedback
- Hardware Integration
- Product Promotion

3.3 Story Telling

HAPPY

A Disabled individual ,unable to use a traditional mouse or keyboard. Uses gesture recognition technology to control their home's lighting system. With just a simple hand gesture , they can turn on lights , open doors , and even control their t.v. This system has improved their independence and quality of life.

SAD

An elderly person , unfair with new technology , struggles to get the gesture recognition system to recognize their hand inovements . they grow frustrated as the lights flicker on and off randomly , cousing stress and making them feel incompetent with technology.

3.3 snapshot of empty mapping

Design For

Date

Design By

Version

USER

- Factory Workers
- Resurches Developers
- Consumers Audience
- Hazardous Environment

STAKEHOLDERS

- System Investors
- Service Providers
- Medical Manufactures
- End Users

ACTIVITIES

- Model Training
- Environment Testing
- Design Improvement
- User Feed back
- Hardware Integration
- Product Promotion

STORY BOARDING

HAPPY → A disabled individual, unable to use a traditional mouse or keyboard, uses gesture recognition technology to control their home's lighting system. with just a simple hand gesture, they can turn on lights, open doors, and even control their TV. This system has dramatically improved their independence and quality of life.

HAPPY → In a high-tech factory, an engineer uses gesture recognition to control heavy machinery from a safe distance. By using simple hand gestures, they can operate the system without touching any controls, increasing efficiency and safety. The engineer feels empowered by the smooth, seamless interaction.

SAD → An elderly person, unfamiliar with new technology, struggles to get the gesture recognition system to recognize their hand movements. They grow frustrated as the lights flicker on and off randomly, causing stress and making them feel incompetent with technology.

SAD → A surgeon tries to use gesture control to switch between MRI scans during a crucial surgery, but the system misreads their gesture. The delay causes unnecessary stress and disrupts the surgery flow, forcing them to switch back to traditional methods. leading to disappointment with the technology.

4. IDEATION CANVAS

Ideation is the process where you generate ideas and solutions through sessions such as sketching, prototyping, brainstorming, brain writing, worst possible idea and a wealth of other ideation techniques.

"Ideation is the mode of design process in which you concentrate on idea generation."

This canvas is designed to be used in the creative ideation stage of a design process, when you are looking to generate ideas for solutions.

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4.1 activities

- gesture detection
- data training
- Signal peocessing
- Modal deployment
- Control execution
- System cutibration

4.2 Understanding about Situation/Context and Location

- Smart home
- Public spaces
- Virtual reality
- Auto motive control
- Assistive technology

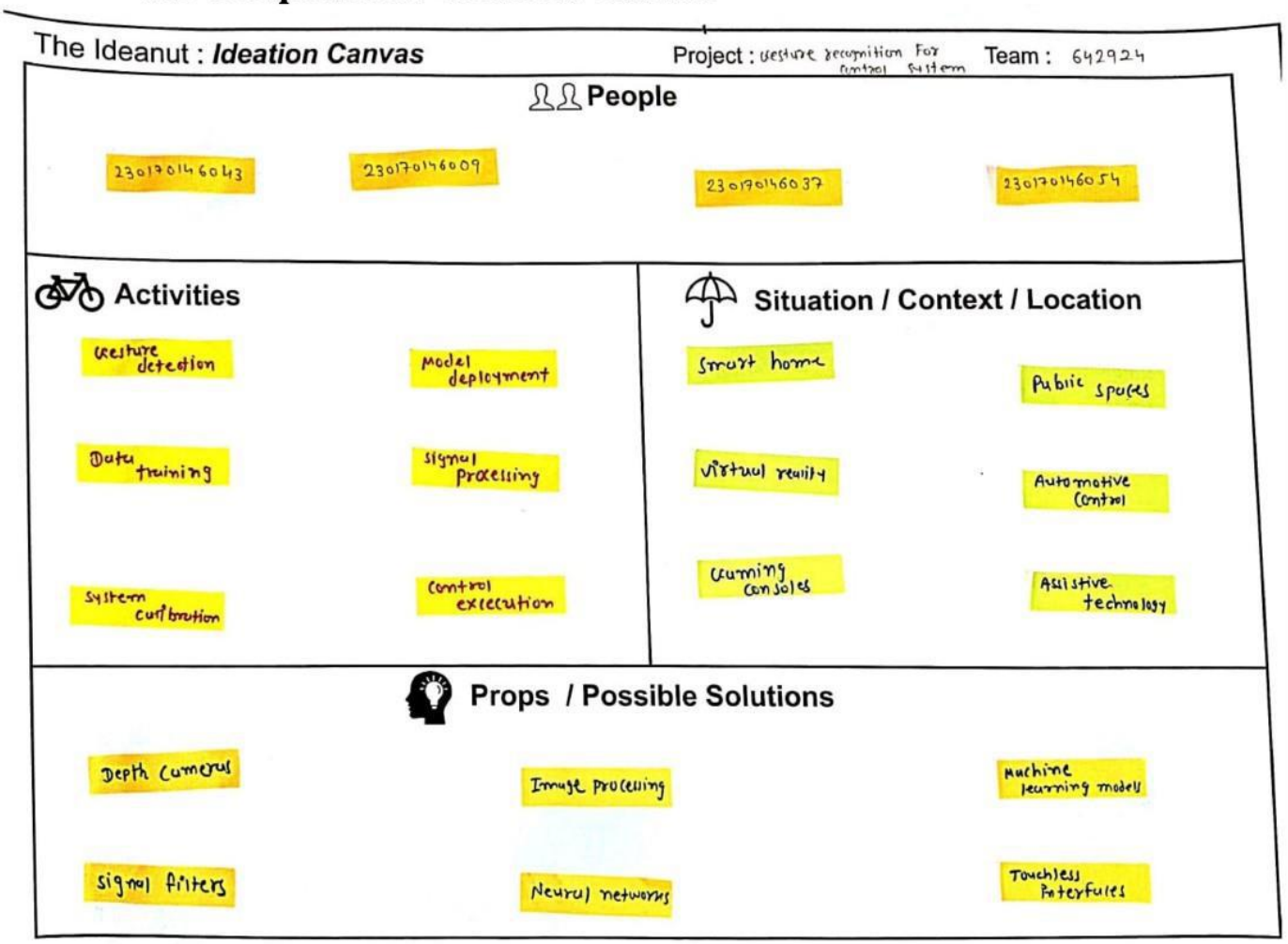
4.3 props

- Depth camers
- Image processing
- Machine hearning models
- Neural networkers
- Touchless interfales

4.4 people

HOD
Faculty
Student
Admin

4.5 Snapshot of Ideation Canvas



5) PRODUCT DEVELOPMENT CANVAS

The Product Development Canvas is often abbreviated as PDC. There are several sections that make up the PDC including user, purpose, experience, imagine and visualize the feelings expectations and experience that users will have after interacting with the developed product and think about the functions, features, components and validation and revalidation.

5.1 description of final problem statement

The final problem statement for "Gesture Recognition for Control Systems" focuses on developing a system that interprets human gestures to control devices or applications. The system will accurately detect and classify hand movements in real-time, enabling seamless, touch-free control. This improves accessibility and user interaction with technology across various applications.

5.2 description of problem solution

The solution involves designing a gesture recognition system using computer vision and machine learning algorithms to detect and classify hand gestures. This system will translate these gestures into commands for controlling various devices or software applications in real time.

5.3 explanation of product components,function,features.

Function

- command mapping
- ML adapbation

Components

- Motion sensors
- Gesture data
- Motion sensors
- Motion gesture

Features

- Read time learning
- Fat response
- ML feedback

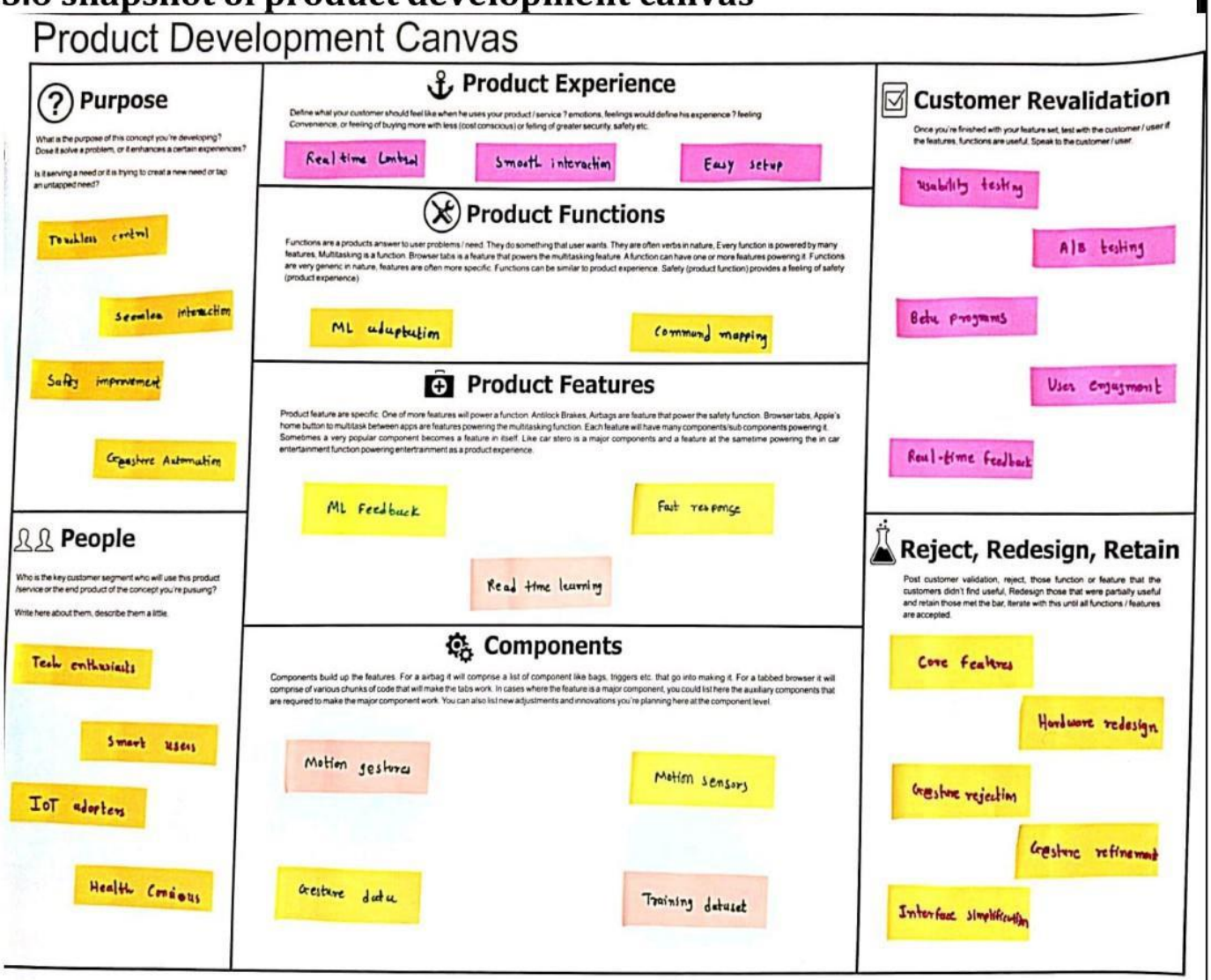
5.4 product experience

- Real time control
- Smooth interaction
- Easy setup

5.5 customer revalidation

- Usability testing
- A/B testing
- Beta programs
- User engagement
- Real – time feedback

5.6 snapshot of product development canvas



6) AEIOU CANVAS

6.1 Briefing to AEIOU

AEIOU basically stands for Activities, Environments, Interactions, Objects, and Users.

This is essentially a Framework to structure all observations you make on the field during user research. But to best understand it, let's continue with the example of the Campus Information Desk. You would realize by now that the search for the efficient solution is a design thinking workshop in itself.

Activities are goal directed sets of actions—paths towards things people want to accomplish.

Environments detail the context and characteristics of the space where activities are being observed.

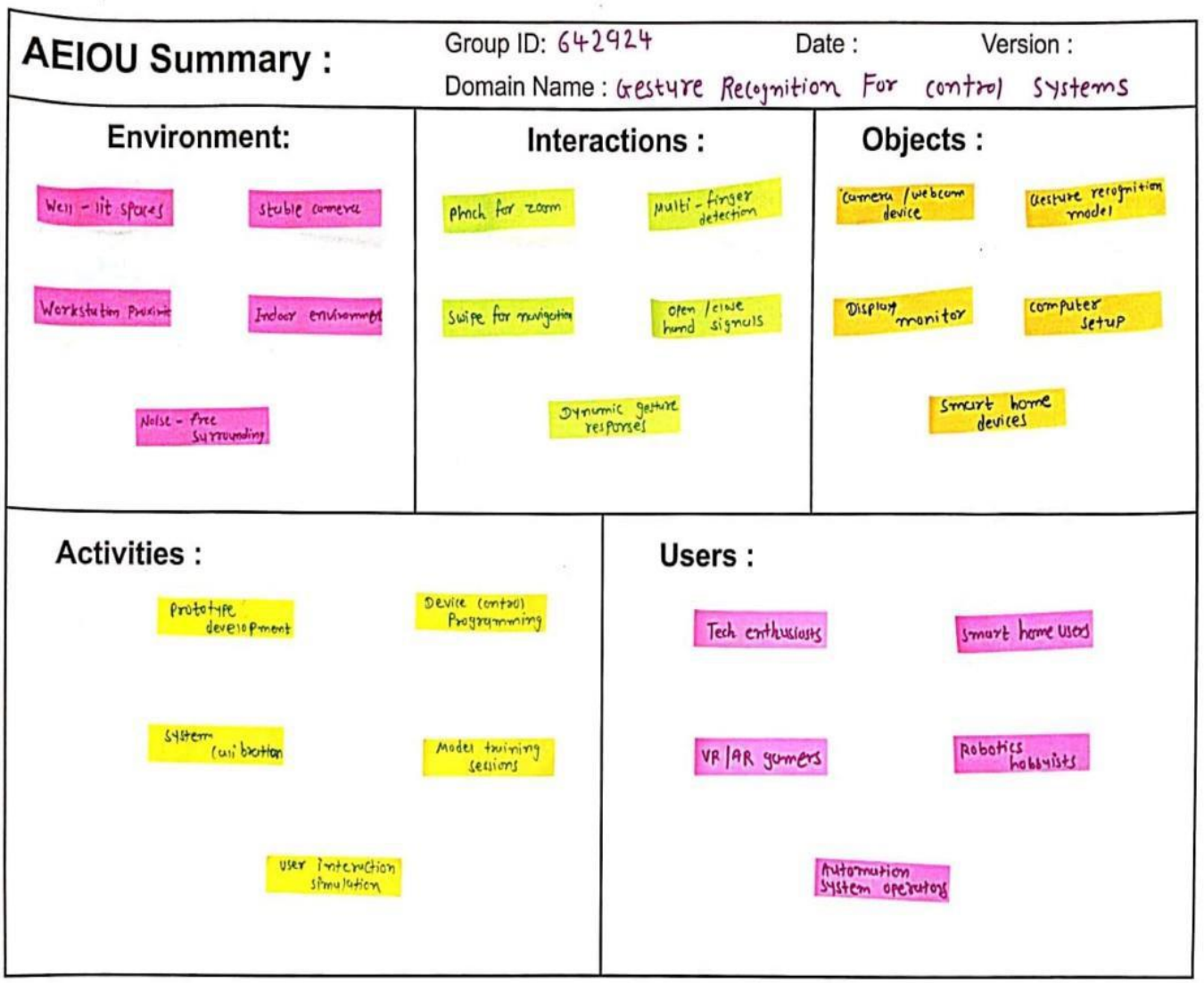
Interactions are between a person and someone or something else; they are the building blocks of activities.

Objects are the building blocks of the environment, key elements sometimes put to complex or even unintended uses, possibly changing their function, meaning and context.

Users who are the people being observed? What are their personalities like?
How do they engage with other people to reach goals?

The AEIOU canvas is a valuable tool in the early stages of design thinking and user-centred design processes. It encourages a holistic understanding of the users and their environment, fostering empathy and insight. Designers and teams can use the information collected through the AEIOU canvas to inform further stages of the design process, such as ideation, prototyping, and testing. The canvas can be adapted and customized based on the specific needs of a project or the nature of the design challenge at hand.

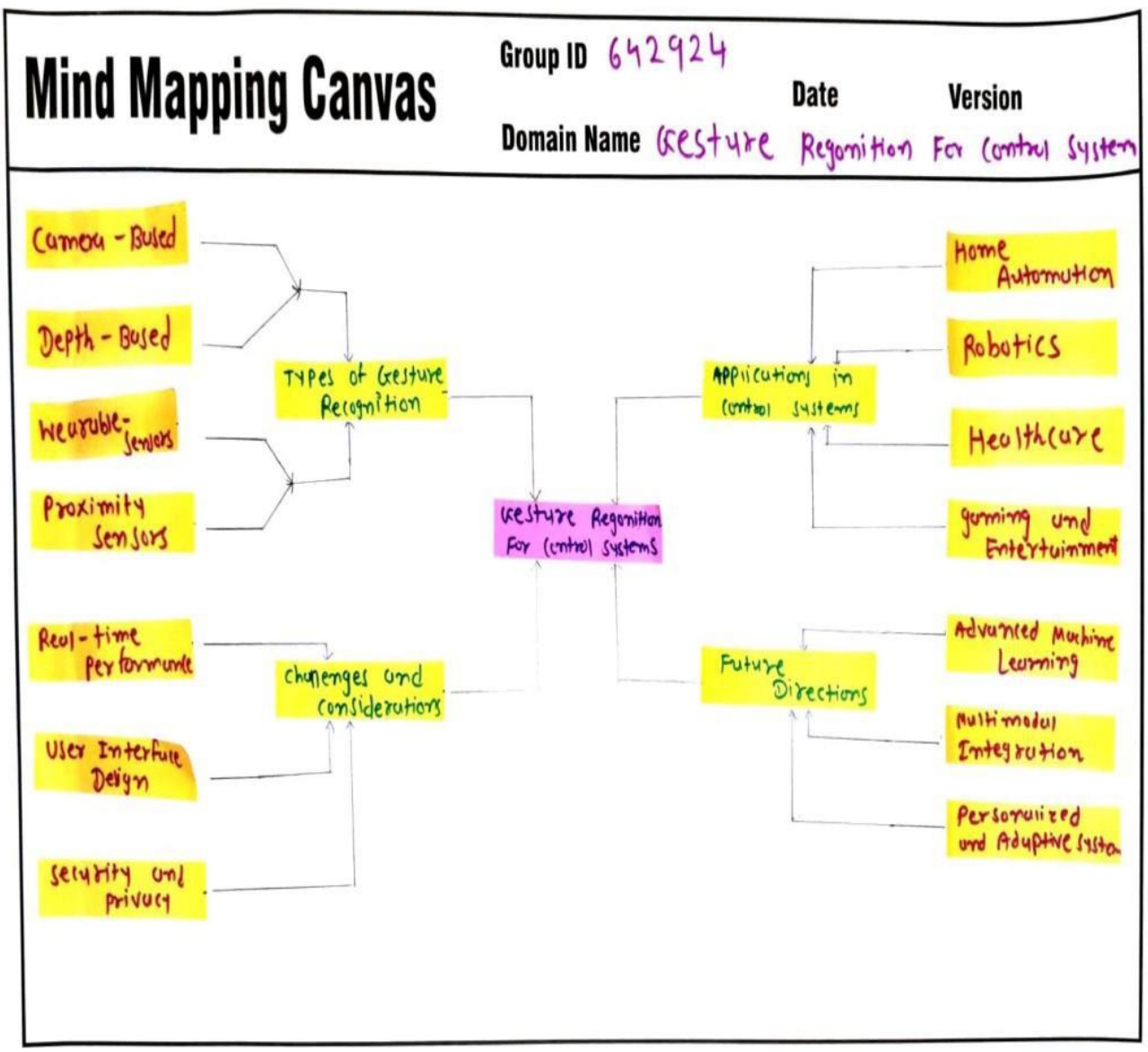
6.2 snapshot of AEIOU canvas



7) MIND MAPPING

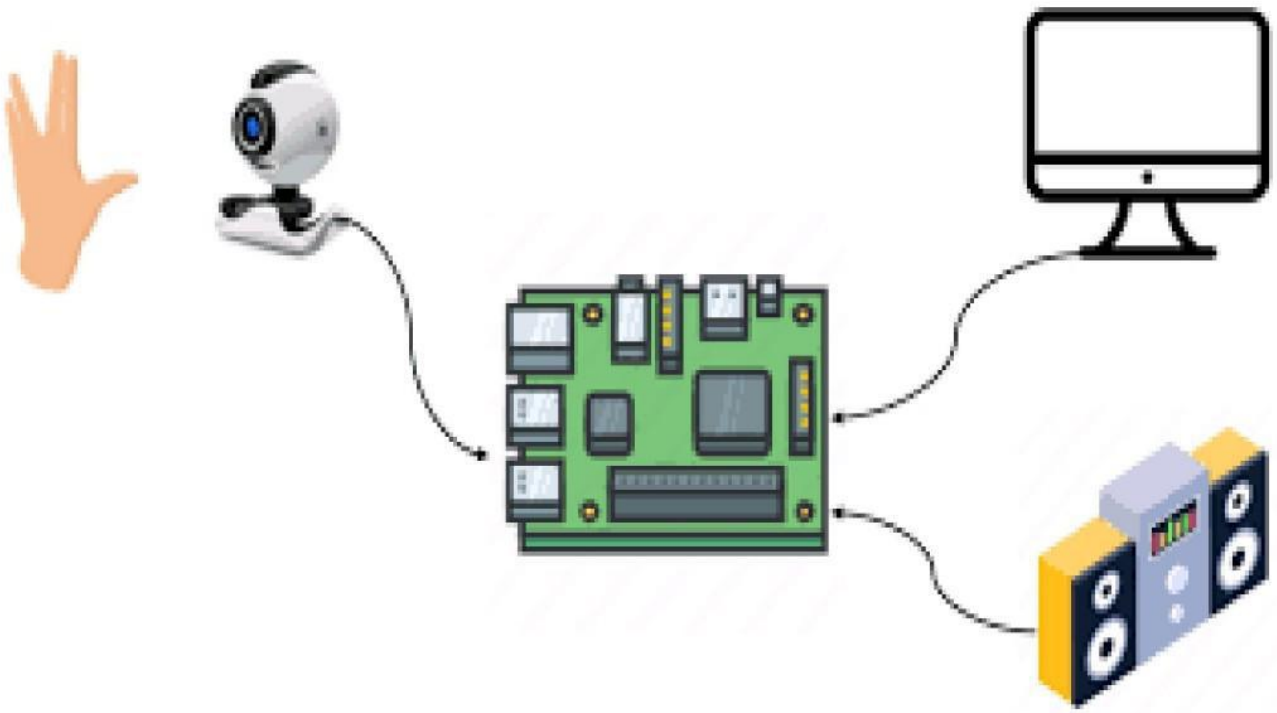
Mind map is a diagram in which information is represented visually, usually with a Central idea placed in the middle and associated ideas arranged around it. Mind mapping is a Powerful technique to help you visually develop and organize ideas and information.

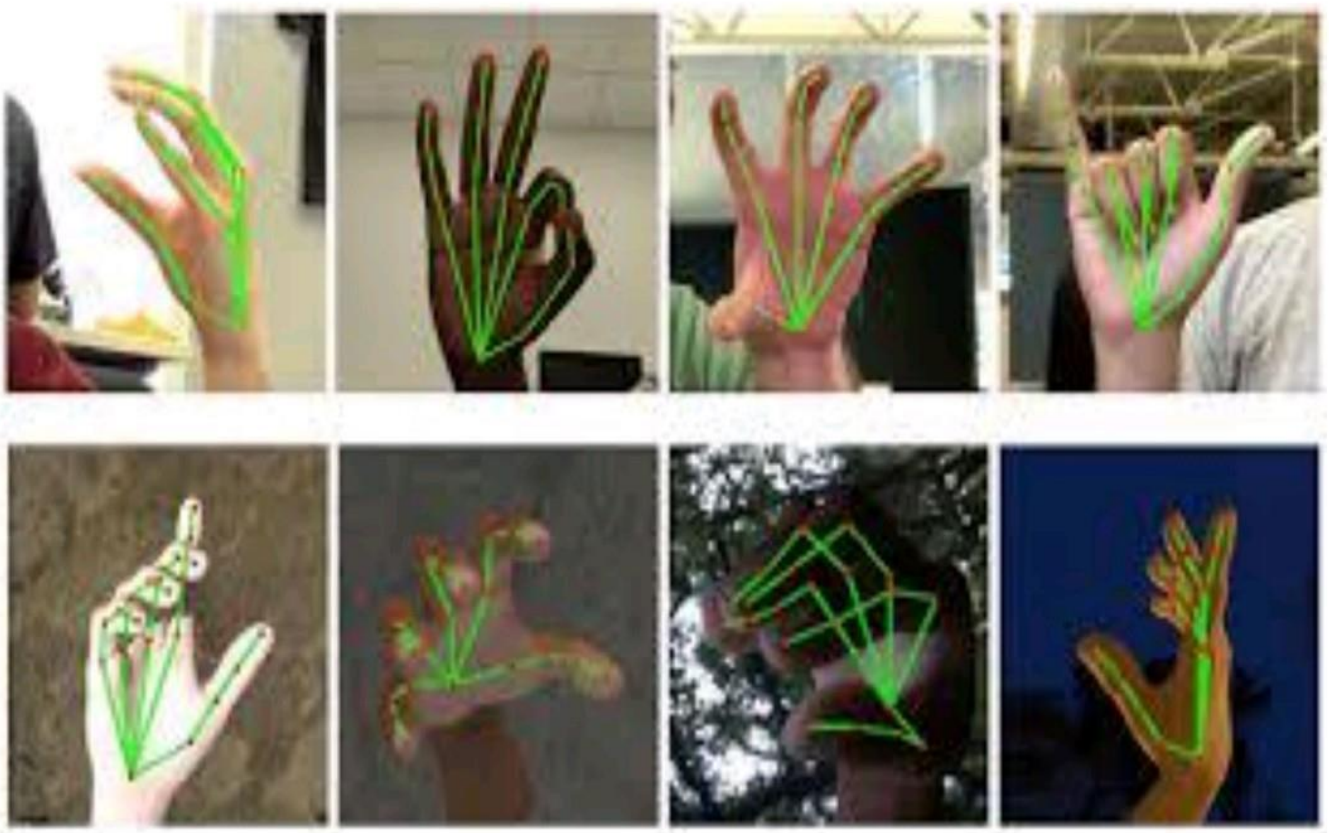
7.1 Snapshot of MIND MAPPING



8) PROTOTYPE CANVAS

8.1 SNAPSHOT OF PROTOTYPE





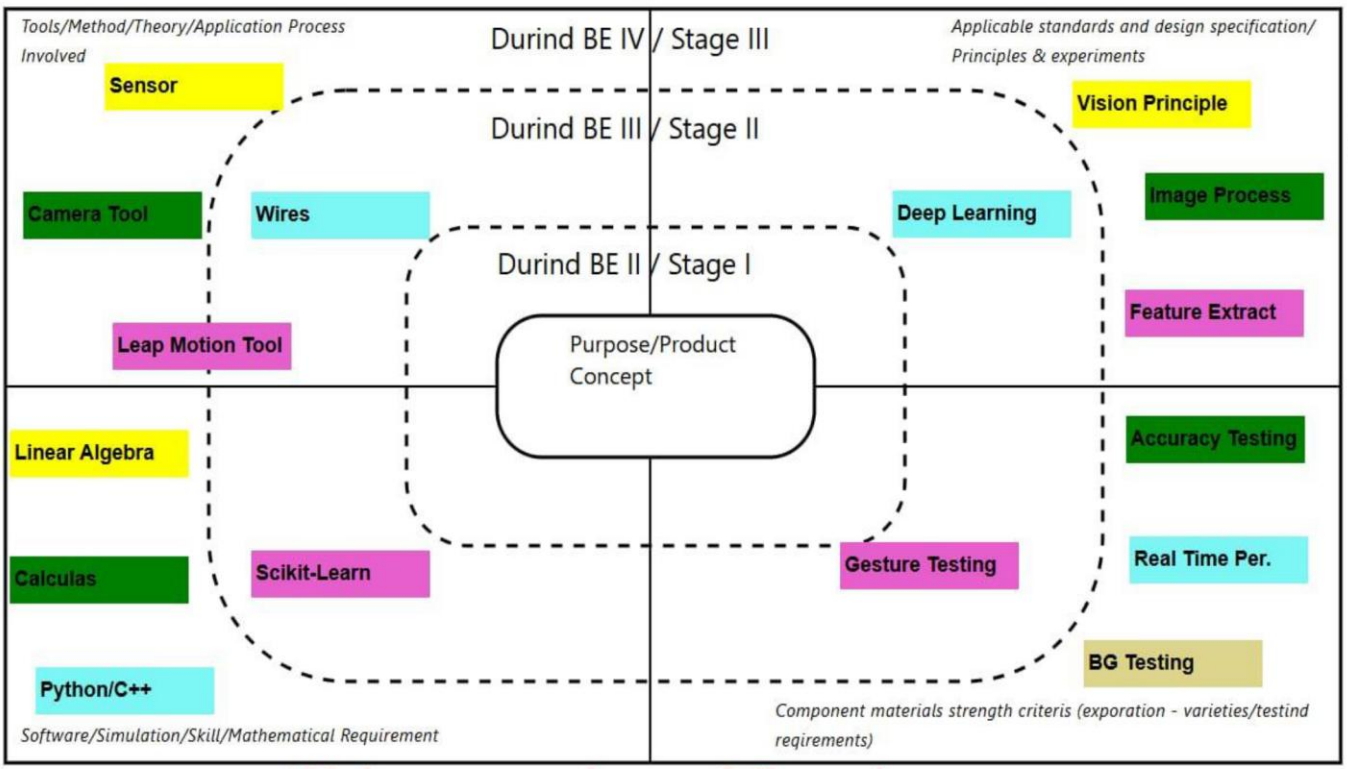
Gesture recognition for control system

Team Id.: 642924

Lerning Need Matrix

Group ID :

Date :



10)CONCLUSION

Gesture recognition as a control system technology represents a significant leap in the field of interactive design. By focusing on user-centered design principles and leveraging advancements in computer vision and AI, gesture-controlled systems can be effectively integrated into everyday applications, enhancing functionality, efficiency, and user experience in various fields from automation to entertainment.

In a Design Engineering project, implementing gesture recognition not only encourages innovation but also aligns with modern trends toward more immersive and natural user interfaces.